Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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This document reports, in full, four robustness checks summarized in the main text, and the variants of the candidate measure (Section S5). All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption.

Table S1: Panel-size recoverability under alternative correlation bars

The main text calls a panel size "recovering" the full-panel ranking once mean Spearman ρ against the full panel exceeds 0.90, a convention borrowed from reliability research's "excellent agreement" bar rather than derived from the data (Methods, Results). Table S1 reports p^* (the smallest panel size reaching the bar) at four alternative bars, for both sparse-coverage datasets.

Table S1: Panel-size p^* (kinases) at four correlation bars. (`pstar_bar_sensitivity.py`)

Klaeger (343 kinases)	entropy	gini	s_score	ratio
bar = 0.80	80	260	50	320
bar = 0.85	80	260	80	320
bar = 0.90	110	290	110	343
bar = 0.95	140	320	200	343

Metz (172 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

The ratio remains the slowest to recover for every bar tested, and the Gini coefficient the second-slowest on Klaeger (on Metz it ties the entropy at every bar); only the relative order of entropy and the S-score is bar-dependent, and only within $[0.80, 0.95]$.

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets to check whether they behave as additional members of the ratio family (WS, RS) or the distribution-based family (CATDS), as claimed in the Discussion. (`bosc_catds_comparison.py`)

Table S2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.804	0.878	0.509	0.955 / 0.960
Klaeger	222	0.909	0.892	0.681	0.850 / 0.985
Anastassiadis	178	0.959	0.891	0.696	0.948 / 0.965
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.80$–0.96), supporting their classification alongside $R(k)$. The CATDS proxy instead correlates more strongly with entropy and Gini than with the ratio on every dataset, supporting the Discussion's claim that CATDS is a distributional construction despite its historical classification as a ratio measure.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

Table S3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.417	0.952	0.456
entropy(literal)	0.417	1.000	0.424	0.336
Gini	0.952	0.424	1.000	0.343
ratio	0.456	0.336	0.343	1.000

Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.570	0.722	0.223
entropy(literal)	0.570	1.000	0.688	0.789
Gini	0.722	0.688	1.000	0.616
ratio	0.223	0.789	0.616	1.000

Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.239	0.879	0.211
entropy(literal)	0.239	1.000	0.416	0.758
Gini	0.879	0.416	1.000	0.229
ratio	0.211	0.758	0.229	1.000

Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.880	0.345	0.489
entropy	0.880	1.000	0.632	0.529
Gini(native)	0.345	0.632	1.000	0.314
ratio	0.489	0.529	0.314	1.000

On Klaeger and Metz, literal entropy correlates more strongly with the ratio than with either S-score or Gini – under this operationalization, entropy would not be classified in the distribution-based family at all on either sparse-coverage dataset. On Davis, literal entropy decouples from both families (consistent with Figure 1 of the main text). The native-Gini check on Anastassiadis

preserves the pattern only partly: the native Gini correlates with the entropy (0.63) but weakly and about equally with the S-score (0.35) and the ratio (0.31).

Table S4: Candidate measure cross-dataset validation

Orientation (correlation with the number of active kinases; more negative is more correctly oriented toward "fewer active targets = more selective"); D3 as defined in Methods, i.e. the percentage of pairs of scored compounds whose order reverses somewhere in $B = [x_0, \tau^*]$, for the emphasis parameter β with the gate x_0 fixed (the featured measure), for the gate x_0 itself swept as a free parameter (the like-for-like comparison with the entropy's baseline; Figure 6 of the main text), and for the entropy's own baseline; and the percentage of compounds for which D4 holds when one kinase is added at the reliability threshold τ^*, the hardest case. B is $[5.0, 6.0]$ on Davis and Klaeger, $[4.0, 6.0]$ on Metz, and $[20, 50]$% inhibition on Anastassiadis; the smoothing width is $T = \tau^* - x_0$ on every dataset (Methods). D3 passes at $\varepsilon = 1\%$. (`candidate_validation.py`, `d3_pair_reversals.py`)

Table S4: Candidate measure: orientation, D3 (pairs reversing in B, %), and D4, across all four datasets.

Dataset	n_{cpd}	orientation r	D3, gate fixed	D3, gate swept	D3, entropy	D4 (%)
Davis	68	−0.969	0.48	8.52	8.82	100.0
Klaeger	222	−0.936	0.54	10.57	11.20	100.0
Anastassiadis	178	−0.940	0.43	12.49	10.27	100.0
Metz	704	−0.785	0.72	22.45	22.60	100.0

The "gate swept" and "entropy" columns confirm, on every dataset, that the candidate's D3 satisfaction rests on fixing the gate x_0 by declaration rather than on the measure being inherently insensitive to its placement: swept over the same interval, the candidate's gate reverses about as many pairs as the entropy's baseline does, while the emphasis parameter, with the gate fixed, reverses fewer than one pair in a hundred. Removing the gate inverts the measure's orientation entirely (gated $r = -0.936$ vs. ungated $r = +0.943$ against n_{active} on Klaeger), which is why the gate, not merely the smooth emphasis, is required for D1 and correct orientation jointly.

S5: Variants of the candidate measure

The featured candidate is one member of a family of measures. Three axes of variation are reported here: the floor, the Rényi order, and the smoothing width T. Each variant is checked against D1–D4 as in the main text; D4 is proved only for the featured measure (Shannon order, $\beta = x_0$, $T \geq 0.58\,(\tau^* - x_0)$), and is otherwise checked by adding one kinase at τ^* to every compound. (`family_construction.py`, `family_panelsize.py`, `candidate_validation.py`)

Quantile floor. Replacing the fixed floor x_0 with a per-compound low quantile of the profile (the 10th percentile) gives a variant that is invariant to shifts of the readout. On Davis and Klaeger it ranks compounds exactly as the featured measure does (Spearman 1.000), no pair of scored compounds reverses as the quantile is varied from the 5th to the 15th percentile, and D4 holds for every compound.

Rényi order. Generalizing the Shannon functional to Rényi order $\alpha > 0$,

$$-H_\alpha(p) = \frac{1}{\alpha - 1} \log_2 \sum_i p_i^\alpha,$$

gives a family that recovers the featured measure at $\alpha = 1$. D1 holds for every order, since the gate is applied before the entropy is computed, and D2 holds vacuously. D4 holds for additions at or below the floor, because $p_i^\alpha = 0$ whenever $p_i = 0$, and holds on every compound with one kinase added at τ^* for each order tested. D3 separates the orders (Table S5): $\alpha \leq 1$ passes on both datasets, $\alpha = 2$ is borderline (it passes on Davis and narrowly fails on Klaeger), and $\alpha = 3$ fails on both.

Table S5: Rényi-order variants on Davis and Klaeger: D3 (pairs reversing in B, %), agreement with the Shannon ranking (Spearman), and panel size p^* on Klaeger (smallest panel reaching mean $\rho > 0.90$; on Metz every order reaches $p^* = 80$). (`family_construction.py`, `family_panelsize.py`)

Order	D3, Davis	D3, Klaeger	Spearman vs. $\alpha = 1$	p^*, Klaeger
$\alpha = 0.5$	0.61	0.13	0.999–1.000	80
$\alpha = 1$	0.48	0.54	—	110
$\alpha = 2$	0.92	1.09	0.999	290
$\alpha = 3$	1.80	1.68	0.997–0.998	290

The order affects panel recoverability on the sparser Klaeger panel. Every order below 1 tested (0.01, 0.25, 0.5, 0.75) reaches $p^* = 80$, ahead of the Shannon order's 110, while $\alpha \in \{2, 3\}$ needs 290, matching the Gini. On Metz every order from 0.01 to 3 reaches $p^* = 80$. This speed does not come from discarding information: the number of distinct scores and the size of the largest tied group are identical across every order tested, including $\alpha \to \infty$, and the only tied compounds are those with at most one active kinase. It reflects how each limit behaves under subsampling. As $\alpha \to 0$ the score approaches a count of the active set, which a random subpanel estimates with low variance; as $\alpha \to \infty$ it depends on the single largest weight, and the estimate is far off whenever the subpanel misses the kinase carrying it. Agreement with the Shannon ranking is as high on the sparsest compounds ($n_{\text{active}} < 5$) as on the rest for $\alpha \leq 0.5$ (≥ 0.98 in both subsets, on Klaeger and Metz), whereas $\alpha \to \infty$ loses agreement most sharply among the sparsest Metz compounds (0.62, against 0.92 for the rest; on Klaeger, 0.98 and 0.95). We feature $\alpha = 1$ because it is the

Shannon entropy, for which D4 is proved; a lower order is a documented alternative with faster panel recoverability on Klaeger.

Smoothing width. The featured measure uses $T = \tau^* - x_0$. On Klaeger, over $T = 0.5$, 0.75, 1, 1.5, and 2 (in units of $\tau^* - x_0 = 1$), orientation is unchanged ($r \approx -0.94$) and the rankings agree closely across all five values (worst-case pairwise Spearman 0.999). D3 passes for $T \geq 0.75$ (pairs reversing as the emphasis baseline is swept: 0.93%, 0.54%, 0.17%, 0.04% for $T = 0.75$, 1, 1.5, 2) and narrowly fails at $T = 0.5$ (1.94%), which also falls below the D4 bound $T \geq 0.58$. As $T \to 0$ the softplus converges to the hard cutoff, recovering the entropy together with its D3 and D4 failures.